# Disease Metabolic Network Modelling:

Separating Numerical Noise from Thermodynamic Inconsistency in Cycle-Consistency Tests on Metabolic Circuits

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**Abstract**

**Background.** A metabolic network at steady state can be represented as an electrical circuit in which Kirchhoff's voltage law corresponds to the Wegscheider condition: the sum of chemical-potential differences around any cycle of reactions is identically zero, and a nonzero cycle sum therefore indicates thermodynamic inconsistency of the reaction data. Any such test requires a tolerance $\varepsilon$ below which a cycle sum is treated as zero. In current practice the tolerance is a fixed round number regarded as a solver setting.

**Methods.** The rounding error of a computed cycle sum was bounded analytically, and the uncertainty of the reaction-level data was propagated through the same sum. Uncertainty of the species potentials was shown not to contribute, because it cancels around every cycle. The sum of the rounding-error bound and the uncertainty threshold defines a cycle-specific tolerance $\varepsilon^*$. The tolerance was evaluated in 29,860 seeded simulation trials and on experimental data: 3,276 measured apparent equilibrium constants from the Thermodynamics of Enzyme-Catalyzed Reactions Database (TECRDB), and a human erythrocyte reaction network assembled from UniProt and Rhea.

**Results.** In IEEE double precision a fixed $\varepsilon = 10^{-6}$ produced no false positives, since the rounding error did not exceed $2.7 \times 10^{-12}$; its deficiency is that it classifies as inconsistent cycle sums that lie within the uncertainty of the data. On measured data all 67 independent cycle sums exceeded $10^{-6}$ kJ/mol. The cycle-specific tolerance classified 7 cycles as inconsistent and 60 as undecidable, the two components of the tolerance differing by 14 orders of magnitude. Under simulated Gaussian noise the false-positive rate of $\varepsilon^*$ was 0.009 at a nominal level of 0.01. Replicate measurements of single reactions, whose cycle sums are zero by construction, showed heavier tails than the Gaussian model assumes; with the threshold calibrated on these replicates, 3 cycles were classified as inconsistent. In the erythrocyte network, pyruvate kinase lies on no cycle and no cycle has a measured equilibrium constant for every reaction, so the test cannot be applied to the most common erythrocyte enzymopathy.

**Conclusion.** The tolerance of a cycle-consistency test is determined by the cycle and by the uncertainty of the data, not by the solver. Computing it replaces unsupported classifications by an explicit statement of what the data can resolve.

**Keywords:** Wegscheider condition; metabolic networks; thermodynamic consistency; cycle basis; floating-point error analysis; detection limit; three-valued classification.

## 1 Introduction

### 1.1 Cycle-consistency tests for metabolic networks

Consider a biochemical reaction network at steady state, in which each species $i$ is assigned a chemical potential $\mu_i$ and each reaction $(i, j)$ a flux $J_{ij}$. If the network is thermodynamically

consistent, the potentials are a well-defined function on the species, and the sum of potential differences around any closed cycle of reactions is zero. In the electrical-circuit representation of reaction networks (Oster, Perelson, & Katchalsky, 1971; Peusner, 1986) this is Kirchhoff's voltage law; expressed in terms of rate constants it is the Wegscheider condition (Schuster & Schuster, 1989; Wegscheider, 1901).

The identity provides a diagnostic test. For each cycle $\ell$ of the network the cycle sum

$$H_\ell \;=\; \sum_{(i,j)\in\ell} (\mu_j - \mu_i) \tag{1}$$

is computed, and the network is classified as inconsistent when some $|H_\ell|$ exceeds a tolerance. Because each sum belongs to a specific cycle, a positive result is localised: it identifies the cycle on which the inconsistency occurs. Global consistency checks, which return a single number for the whole network, do not provide this information.

The outcome of the test depends on the tolerance $\varepsilon$. In common practice $\varepsilon$ is fixed at a value such as $10^{-6}$, $10^{-9}$ or a multiple of the machine epsilon, and is treated as a numerical setting of the solver. The present study shows that this practice has a measurable cost and that an appropriate tolerance is not a constant but a quantity computed for each cycle.

### 1.2 Limitations of a fixed tolerance

For a consistent network, (1) is not approximately but identically zero: around a cycle $i_1 \to \cdots \to i_m \to i_1$ the differences telescope to $\mu_{i_1} - \mu_{i_1}$ for any values of $\mu$. The value returned by a floating-point computation is therefore entirely rounding error and carries no information about the network.

The magnitude of this rounding error is not constant. A sum of $m$ differences of numbers of magnitude up to $\Lambda$ accumulates an absolute error that increases with both $m$ and $\Lambda$ (D. Goldberg, 1991; Higham, 2002). A cycle of length 20 through species with potentials in the range $\pm 2000$ kJ/mol returns a computed sum several orders of magnitude larger than a three-membered cycle through species with potentials in the range $\pm 10$ kJ/mol, although both sums are exactly zero. A single threshold placed between the two error scales classifies the first cycle as inconsistent although it is not; a threshold placed above both fails to detect defects on the second cycle that the arithmetic could have resolved.

The practical severity of this effect depends on the floating-point format. In IEEE-754 double precision (binary64) the rounding error remains at least six orders of magnitude below $\varepsilon = 10^{-6}$, so this threshold never produces a false positive from rounding; the two-sided failure occurs in single precision (binary32) (Section 3.1). The threshold does, however, lie far below the uncertainty of measured reaction data. On measured data every cycle sum exceeds $10^{-6}$ (Section 3.3), and a fixed threshold then classifies all cycles as inconsistent.

### 1.3 A tolerance computed for each cycle

The tolerance is computed here from two sources that are kept separate because they address different questions. The first is numerical: how large can a computed cycle sum be when the exact sum is zero, given the arithmetic and the magnitudes involved (Section 2.2; Higham, 2002)? The second is empirical: the reaction data (reaction Gibbs energies, equilibrium constants or ratios of rate constants) carry uncertainty (Alberty, 1998; Flamholz, Noor, Bar-Even, & Milo, 2012; Noor, Haraldsdóttir, Milo, & Fleming, 2013), and a cycle sum smaller than that uncertainty propagated through the sum cannot be distinguished from zero (Section 2.3). Uncertainty in the species potentials, in contrast, cancels around every cycle. The tolerance is the sum of the two contributions. Because there are two thresholds, the classification has three outcomes: below the rounding-error bound the cycle is consistent; above the full tolerance it is inconsistent; between the two the data supplied do not determine the answer (Theorem 2.27).

### 1.4 Dependence of localisation on the cycle basis

A connected network with $|E|$ reactions and $|V|$ species has $|E| - |V| + 1$ independent cycles, and the choice of these cycles is not unique (Diestel, 2017; Kavitha et al., 2009). If the set of flagged cycles depends on that choice, the localisation provided by the test requires qualification. Section 2.5 shows that the network-level classification is independent of the basis, since a nonzero linear functional cannot vanish on a spanning set (Theorem 2.29), whereas the set of flagged cycles is not: there are networks in which one basis attributes a defect to a single cycle and another to three (Proposition 2.30). The quantity that is independent of the basis is the set of edges common to all flagged cycles, and the localisation is reported in terms of that set.

### 1.5 Contributions and organisation

The following results are established: a bound on the rounding error of a computed cycle sum (Theorem 2.12); the cancellation of species-level uncertainty, so that only reaction-level uncertainty contributes to a cycle sum (Lemma 2.17), and the resulting uncertainty threshold (Theorem 2.19); a cycle-specific tolerance $\varepsilon^*(\ell)$ that satisfies both a false-positive bound (Theorem 2.23) and a detection guarantee (Theorem 2.24), which no fixed tolerance satisfies simultaneously; the necessity of a three-valued classification (Theorem 2.27); the distinction between the basis-independent classification (Theorem 2.29) and the basis-dependent set of flagged cycles (Proposition 2.30 and definition 2.32); and a protocol with pre-registered criteria, implemented in a typed language in which a classification cannot be obtained without its tolerance (Section 2.8). The results are tested in simulation, where the true state is known (Sections 3.1 and 3.2), and on measured equilibrium constants and the human erythrocyte network (Section 3.3).

## 2 Materials and Methods

Section 2.1 introduces the circuit representation and shows that the cycle sum vanishes identically, so that its computed value consists of error alone. Section 2.2 bounds the rounding error, and Section 2.3 propagates the reported thermodynamic uncertainty through the same sum. Section 2.4 combines the two into the cycle-specific tolerance and derives the three-valued classification. Section 2.5 treats the dependence of the result on the choice of cycle basis. Section 2.6 states the consistency check that the reference network must pass before any diagnostic conclusion is drawn, Section 2.7 describes the experimental data and the form of the test they require, and Section 2.8 summarises the typed implementation and the experimental protocol with its pre-registered criteria.

Proofs of all numbered results are given in the Supplementary Material. All computations were performed in Python 3 with NumPy Harris et al. (2020), SciPy Virtanen et al. (2020) and NetworkX Hagberg, Schult, and Swart (2008) on a single workstation, in IEEE-754 binary64 arithmetic except where binary32 is stated, with eQuilibrator (Beber et al., 2022) for the thermodynamic transformations of Section 2.7. The validation suite is seeded (`seed` = 20260819) and writes every reported quantity to a JSON record from which the tables and figures are generated.

**Disclosure of AI use.** Generative artificial intelligence (Anthropic Claude) was used to assist with manuscript preparation, specifically language editing and the drafting of code for the validation suite and figure generation. It was not used to generate research data, results, or scientific conclusions. All mathematical results, their proofs, and all reported measurements were derived and verified by the author, who takes full responsibility for the content of this manuscript.

## 2.1 Circuit representation of a reaction network

### 2.1.1 Species, potentials and fluxes

**Definition 2.1** (Reaction network)**.** A *reaction network* is a pair $(V, E)$ with $V = \{1, \dots, n\}$ indexing molecular species and $E \subseteq V \times V$ a set of directed elementary conversions, referred to as edges or reactions. Each species carries a standard chemical potential $\mu_i^\circ \in \mathbb{R}$ and a concentration $c_i > 0$; each edge carries a rate constant $k_{ij} > 0$. Quantities assigned to edges are referred to as *reaction data*.

**Definition 2.2** (Chemical potential)**.** The chemical potential of species $i$ is

$$\mu_i \;=\; \mu_i^\circ + RT \ln c_i, \tag{2}$$

with $R$ the gas constant and $T$ the absolute temperature. The value $RT = 2.577$ kJ/mol at $T = 310$ K is used throughout (Atkins, de Paula, & Keeler, 2018).

Equation (2) is the ideal-solution form; activity corrections are discussed in Section 3.4.

**Definition 2.3** (Conductance and flux)**.** For $(i, j) \in E$, the *conductance* and the *flux* in the linear-response regime near equilibrium are

$$G_{ij} \;=\; \frac{k_{ij} c_i}{RT}, \qquad J_{ij} \;=\; G_{ij}\,(\mu_i - \mu_j). \tag{3}$$

Equation (3) is Ohm's law with the chemical potential in the role of the electrical potential and the flux in the role of the current (Beard, Liang, & Qian, 2002; Oster et al., 1971; Qian & Beard, 2005). It is the linearisation of mass-action kinetics and holds to first order in $(\mu_i - \mu_j)/RT$. The results below do not depend on it, since cycle sums involve only the potentials; the analogy is used for the correspondence of the two Kirchhoff laws.

**Proposition 2.4** (Kirchhoff correspondence)**.** *Let a reaction network be at steady state. Then*

*(i)* (Current law.) *For every species $i$, $\sum_{j:(i,j)\in E} J_{ij} - \sum_{j:(j,i)\in E} J_{ji} = 0$ if and only if the concentration of $i$ is stationary.*

*(ii)* (Voltage law.) *For every directed cycle $i_1 \to i_2 \to \cdots \to i_m \to i_1$ in $E$, $\sum_{r=1}^{m} (\mu_{i_{r+1}} - \mu_{i_r}) = 0$.*

*Proof.* See Supplementary Material, Section S1.

*Remark* 2.5 (Exactness of the cycle identity). Proposition 2.4(ii) is an algebraic identity, not an approximation: once $\mu$ is defined as a function on $V$, every cycle sum is zero without an error term. The test therefore involves no model error. On exact data a nonzero computed value is either rounding error or evidence that no potential function exists; on uncertain data it must in addition be compared with the propagated uncertainty. Sections 2.2 and 2.3 provide the means of distinguishing these cases.

### 2.1.2 Cycle space and cycle bases

**Definition 2.6** (Cycle space, cycle basis)**.** Let $G = (V, E)$ be connected with $|V| = n$ and $|E| = m$, and orient each edge arbitrarily. The *cycle space* $\mathcal{Z}(G) \subseteq \mathbb{R}^m$ is the kernel of the incidence matrix, the $n \times m$ matrix with entry $-1$ at the tail and $+1$ at the head of each edge. Its dimension is the cyclomatic number $\nu = m - n + 1$. A *cycle basis* is a set of $\nu$ cycles whose incidence vectors span $\mathcal{Z}(G)$ (Bollobás, 1998; Diestel, 2017; Kavitha et al., 2009).

**Definition 2.7** (Cycle sum)**.** For a cycle $\ell$ with incidence vector $z_\ell \in \{-1, 0, 1\}^m$ and potential vector $\mu \in \mathbb{R}^n$, the *cycle sum* is

$$H_\ell(\mu) \;=\; \sum_{e \in E} z_\ell(e)\, \Delta\mu_e \;=\; z_\ell^\top B^\top \mu, \tag{4}$$

where $B$ is the incidence matrix and $\Delta\mu_e = \mu_{\mathrm{head}(e)} - \mu_{\mathrm{tail}(e)}$. The number of edges of $\ell$ is denoted $|\ell|$, and $\Lambda_\ell = \max_{i \in \ell} |\mu_i|$ is the largest potential magnitude on the cycle.

**Lemma 2.8** (Vanishing of cycle sums of potential differences)**.** *For every cycle $\ell$ and every $\mu \in \mathbb{R}^n$, $H_\ell(\mu) = 0$.*

*Proof.* See Supplementary Material, Section S2.

Lemma 2.8 restates Proposition 2.4(ii) in terms of linear algebra: the cycle sum is a linear functional that vanishes on all of $\mathbb{R}^n$. Its computed value therefore reflects the computation and not the network, except when the reaction data are not derived from any potential vector, which is the case the test is designed to detect.

### 2.1.3 Thermodynamic inconsistency of reaction data

**Definition 2.9** (Reaction-level potential differences and defect)**.** Suppose that for each edge $e$ the data supply a potential difference $d_e \in \mathbb{R}$, obtained from measurement or from thermodynamic tables and not necessarily of the form $\Delta\mu_e$ for some $\mu$. The *defect* of the network is the distance of $d$ from the set of potential differences:

$$\delta(d) \;=\; \min_{\mu\in\mathbb{R}^n} \|d - B^\top \mu\|_2. \tag{5}$$

The network is *consistent* if $\delta(d) = 0$, that is, if $d \in \operatorname{im} B^\top$.

**Proposition 2.10** (Characterisation of consistency by cycle sums)**.** *$\delta(d) = 0$ if and only if $H_\ell(d) := z_\ell^\top d = 0$ for every cycle $\ell$, and it suffices to check the cycles of any cycle basis.*

*Proof.* See Supplementary Material, Section S3.

Proposition 2.10 justifies the test. It guarantees that every cycle basis detects a defect, but it does not determine which basis cycle carries the nonzero sum; this question is treated in Section 2.5.

## 2.2 Rounding-error bound

### 2.2.1 Floating-point model

**Assumption 2.11** (Floating-point model)**.** Arithmetic is IEEE-754 binary64 (double precision) (IEEE, 2019) with unit roundoff $u = 2^{-53} \approx 1.11 \times 10^{-16}$, the largest relative error of a single rounding. Each elementary operation satisfies $\mathrm{fl}(a \operatorname{op} b) = (a \operatorname{op} b)(1+\theta)$ with $|\theta| \le u$, and no overflow or underflow occurs (D. Goldberg, 1991; Higham, 2002).

The standard notation of rounding-error analysis is used: if $|\theta_i| \le u$ and $ku < 1$, then $\prod_{i=1}^{k}(1+\theta_i)^{\pm 1} = 1 + \vartheta_k$ with

$$|\vartheta_k| \;\le\; \gamma_k \;:=\; \frac{ku}{1-ku}. \tag{6}$$

### 2.2.2 Error bound for a computed cycle sum

**Theorem 2.12** (Rounding-error bound for a cycle sum)**.** *Let $\ell$ be a cycle of length $L = |\ell|$ through species with potentials bounded by $\Lambda_\ell = \max_{i\in\ell}|\mu_i|$, and let $\widehat{H}_\ell$ denote the value of (4) computed in the arithmetic of Assumption 2.11 by forming the $L$ differences and summing them in order. Then, although $H_\ell = 0$ exactly,*

$$\bigl|\widehat{H}_\ell\bigr| \;\le\; \gamma_{2L}\cdot 2L\Lambda_\ell \;=\; \frac{2Lu}{1-2Lu}\cdot 2L\Lambda_\ell \;=\; O\Bigl(L^2 u\Lambda_\ell\Bigr). \tag{7}$$

*If the differences are accumulated with compensated (Kahan) summation, which carries a correction term for the rounding error of each addition, the bound reduces to $O(Lu\Lambda_\ell)$.*

*Proof.* See Supplementary Material, Section S4.

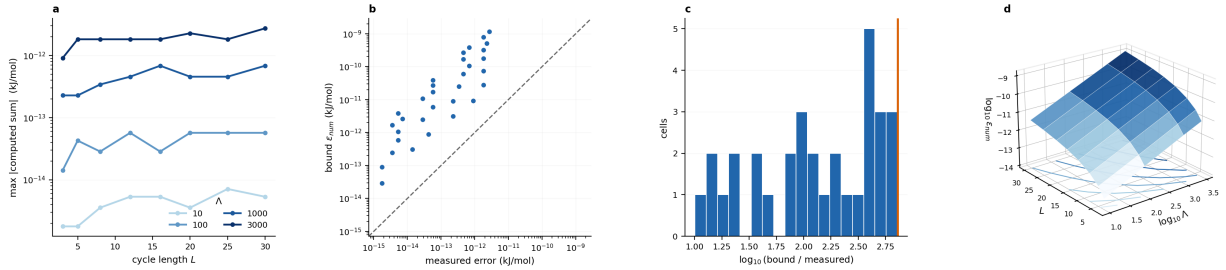



Figure 1: **Rounding error of a computed cycle sum.** All quantities are measured on thermodynamically consistent circuits, whose cycle sum is identically zero (Lemma 2.8), so that the computed value equals the accumulated rounding error. IEEE-754 binary64, $u = 2^{-53}$. (**A**) Maximum computed $|\widehat{H}_\ell|$ over 60 random ring circuits per grid cell, against cycle length $L \in \{3, 5, 8, 12, 16, 20, 25, 30\}$, one curve per potential range $\Lambda \in \{10, 10^2, 10^3, 3 \times 10^3\}$ kJ/mol. The error increases with both parameters over three orders of magnitude, from $1.78 \times 10^{-15}$ to $2.73 \times 10^{-12}$, and its maximum remains six orders of magnitude below the conventional threshold $10^{-6}$ (Remark 3.1). (**B**) Bound $\varepsilon_{\mathrm{num}}(\ell) = \gamma_{2L} \cdot 2L\Lambda_\ell$ against measured error for all 32 cells; the dashed diagonal marks equality. All points lie above it, with no violation in 1920 trials. (**C**) Overestimation factor of the bound: histogram over the 32 cells of the mean bound divided by the maximum measured error (10.1 to 722); the orange line marks the median over individual trials, 729. The bound exceeds the observed error by one to three orders of magnitude, increasingly with $L$, because it grows as $L^2$ while random rounding errors accumulate more slowly (Higham, 2002). (**D**) $\log_{10} \varepsilon_{\mathrm{num}}$ as a surface over $(\log_{10} \Lambda, L)$, ranging from $2.93 \times 10^{-14}$ to $1.17 \times 10^{-9}$ kJ/mol. The bound increases in both arguments, so it must be computed for each cycle.

**Corollary 2.13** (Dependence of the bound on length and potential range)**.** *Let $\ell_1, \ell_2$ be cycles with $L_1\Lambda_1 \neq L_2\Lambda_2$. Then the bounds of (7) differ by the factor $(L_1\Lambda_1)/(L_2\Lambda_2)$, up to the ratio of the $\gamma$ terms, and no constant $\varepsilon$ lies above the first and below the second whenever this ratio exceeds the dynamic range required of the test.*

*Proof.* See Supplementary Material, Section S5.

*Remark* 2.14 (Potential magnitude and potential spread). The bound is stated in terms of $\Lambda_\ell = \max|\mu_i|$ because every difference then satisfies $|\Delta\mu_e| \leq 2\Lambda_\ell$. The actual rounding error is governed by a different quantity. A floating-point subtraction is rounded relative to its result, and by Sterbenz's lemma it is exact when the two operands lie within a factor of two of each other (D. Goldberg, 1991; Higham, 2002). The error is therefore determined by the spread $\max \mu - \min \mu$ on the cycle rather than by the magnitude of the potentials. Standard chemical potentials in metabolic reconstructions are negative numbers of magnitude $10^2$–$10^3$ kJ/mol (Alberty, 1998; Noor et al., 2013), whose spread on a cycle is often much smaller, and the bound in $\max|\mu|$ then overestimates the error by the ratio of the two. A common shift of all potentials changes neither a cycle sum nor its rounding error but does change $\Lambda_\ell$; Section 2.2.3 uses this to reduce the bound.

### 2.2.3 Reducing the bound by centring the potentials

**Proposition 2.15** (Invariance under a common potential shift)**.** *For any $c \in \mathbb{R}$, replacing $\mu$ by $\mu + c\mathbf{1}$ leaves every cycle sum unchanged but changes the bound (7) through $\Lambda_\ell$. In the terminology of physics, the choice of $c$ is a gauge choice. The value minimising the bound on cycle $\ell$ is $c^* = -\frac{1}{2}(\max_{i\in\ell} \mu_i + \min_{i\in\ell} \mu_i)$, which gives $\Lambda_\ell = \frac{1}{2}(\max_{i\in\ell} \mu_i - \min_{i\in\ell} \mu_i)$, half the spread.*

*Proof.* See Supplementary Material, Section S6.

*Remark* 2.16 (Effect of centring on the bound and on the error). For a cycle through species with potentials $-1750 \pm 30$ kJ/mol, centring reduces $\Lambda_\ell$ from 1780 to 30 and the bound by a factor of approximately 60. The rounding error itself is unchanged. Every potential on such a cycle lies

within a factor of two of every other, so by Sterbenz's lemma every difference and every partial sum is exact, and the computed cycle sum is exactly zero with or without centring. Centring requires one pass over the cycle and changes no reported quantity; it reduces the overestimation of the bound, not the error of the computation. It is applied throughout, and both components of the tolerance are reported in the protocol (Supplementary Section S16).

## 2.3 Uncertainty threshold

### 2.3.1 Propagation of uncertainty to a cycle sum

The rounding-error bound of Section 2.2 applies to exact reaction data. Measured and estimated data are not exact. Standard Gibbs energies estimated by group- or component-contribution methods carry reported uncertainties of several kJ/mol (Alberty, 1998; Flamholz et al., 2012; Noor et al., 2013), and concentrations carry measurement uncertainty. Only part of this uncertainty affects a cycle sum.

**Lemma 2.17** (Cancellation of species-level errors)**.** *Let the reaction data be differences of estimated species potentials, $d = B^{\top}\hat{\mu}$ with $\hat{\mu}_i = \mu_i + \eta_i$ for arbitrary errors $\eta_i$. Then $H_\ell(d) = 0$ for every cycle $\ell$ and every $\eta$.*

*Proof.* See Supplementary Material, Section S7.

Concentrations enter each reaction datum through the terms $RT \ln c_i$ of its two endpoints, so concentration uncertainty does not reach a cycle sum. The same applies to the uncertainty of standard Gibbs energies of formation when reaction Gibbs energies are computed from them (Noor et al., 2013). Reaction data constructed in this way are consistent by construction, and a cycle-sum test applied to them detects only errors of bookkeeping. Thermodynamic inconsistency arises from reaction data that are not derived from a single potential vector: reaction Gibbs energies or equilibrium constants measured or annotated reaction by reaction, or ratios of forward and reverse rate constants, which the Haldane relation links to the standard reaction Gibbs energy (Schuster & Schuster, 1989). A model of uncertainty attached to species potentials therefore predicts no variation of the cycle sum, as the simulations of Section 3.2 confirm.

**Definition 2.18** (Uncertainty of a reaction datum)**.** Each reaction datum $d_e$ carries an uncertainty $\sigma_e > 0$, the reported standard error of the part of $d_e$ that is estimated independently for that reaction.

**Theorem 2.19** (Uncertainty threshold for a cycle sum)**.** *Let $d_e = \bar{d}_e + \eta_e$ with $\bar{d}$ consistent and the errors $\eta_e$ independent with standard deviations $\sigma_e$. Then the cycle sum $H_\ell(d)$ has mean zero and standard deviation*

$$s_\ell \;=\; \Big(\textstyle\sum_{e\in E} z_\ell(e)^2\,\sigma_e^2\Big)^{1/2}, \tag{8}$$

*which for a simple cycle is $s_\ell = (\sum_{e\in\ell}\sigma_e^2)^{1/2}$. With*

$$\varepsilon_{\mathrm{data}}(\ell) \;=\; z_{1-\alpha/2}\; s_\ell, \tag{9}$$

*where $z_{1-\alpha/2}$ is the standard normal quantile, a consistent network satisfies $|H_\ell| \le \varepsilon_{\mathrm{data}}(\ell)$ with probability $1-\alpha$ when the $\eta_e$ are normally distributed.*

*Proof.* See Supplementary Material, Section S7.

*Remark* 2.20 (Reaction data without reported uncertainty). Many reconstructions report no uncertainty for individual reactions (Brunk et al., 2018; Orth, Thiele, & Palsson, 2010), and the uncertainty threshold cannot then be computed. No default value is substituted, since an assumed $\sigma$ would turn a question the data cannot answer into an apparently definite answer. The protocol of Section 2.8 instead reports $\varepsilon_{\mathrm{data}}$ as undefined and assigns the intermediate class of Theorem 2.27 to every cycle whose sum exceeds $\varepsilon_{\mathrm{num}}$.

## 2.4 Cycle-specific tolerance

### 2.4.1 Definition and error guarantees

**Definition 2.21** (Cycle-specific tolerance)**.** For a cycle $\ell$, the *cycle-specific tolerance* is

$$\varepsilon^*(\ell) \;=\; \varepsilon_{\mathrm{num}}(\ell) + \varepsilon_{\mathrm{data}}(\ell), \tag{10}$$

with $\varepsilon_{\mathrm{num}}(\ell)$ the bound (7) of Theorem 2.12, computed after centring (Proposition 2.15), and $\varepsilon_{\mathrm{data}}(\ell)$ the threshold (9) of Theorem 2.19. If no uncertainties are supplied, $\varepsilon_{\mathrm{data}}$ is undefined and Remark 2.20 applies.

*Remark* 2.22 (Sum of the two components). A computed cycle sum contains rounding error and data error together, so the tolerance must cover both. If the maximum of the two components were used in place of their sum, neither Theorem 2.23 nor Theorem 2.24 would hold in the presence of data error. In all settings reported here $\varepsilon_{\mathrm{num}}/\varepsilon_{\mathrm{data}} < 10^{-9}$, so the two definitions agree to the precision reported.

**Theorem 2.23** (False-positive bound)**.** *Let the network be consistent, $\delta(d) = 0$ in the sense of Definition 2.9, and let the potentials be exact. Then for every cycle $\ell$ the computed sum satisfies $|\widehat{H}_\ell| \le \varepsilon^*(\ell)$ deterministically. If the reaction data carry independent errors as in Theorem 2.19, the event $|\widehat{H}_\ell| > \varepsilon^*(\ell)$ has probability at most $\alpha$.*

*Proof.* See Supplementary Material, Section S8.

**Theorem 2.24** (Detection guarantee)**.** *Suppose a defect is present with cycle sum $|H_\ell| = D$ on a cycle $\ell$, and $D > 2\varepsilon^*(\ell)$. If the reaction data are otherwise exact, the computed value satisfies $|\widehat{H}_\ell| > \varepsilon^*(\ell)$ and the test flags $\ell$. If they carry independent errors as in Theorem 2.19, the test flags $\ell$ with probability at least $1 - \alpha$.*

*Proof.* See Supplementary Material, Section S9.

*Remark* 2.25 (Decision threshold and detection limit). Theorems 2.23 and 2.24 correspond to the two limits of detection theory in analytical chemistry (Currie, 1968): $\varepsilon^*$ is the decision threshold (critical level), above which a result is reported as a detection, and $2\varepsilon^*$ is the detection limit, the smallest true defect detected with probability at least $1 - \alpha$. For a defect between $\varepsilon^*$ and $2\varepsilon^*$ detection is not guaranteed. The width of this interval is determined by the precision of the arithmetic and of the data, and it can be reduced only by improving either. A fixed tolerance does not remove the interval; it assigns every value within it to one of the two outcomes.

### 2.4.2 Three-valued classification

**Definition 2.26** (Verdict)**.** For a cycle $\ell$ with computed sum $\widehat{H}_\ell$, the *verdict* is

$$\mathrm{verdict}(\ell) = \begin{cases} \mathsf{CONSISTENT} & |\widehat{H}_\ell| \le \varepsilon_{\mathrm{num}}(\ell), \\ \mathsf{UNDECIDABLE} & \varepsilon_{\mathrm{num}}(\ell) < |\widehat{H}_\ell| \le \varepsilon^*(\ell), \\ \mathsf{INCONSISTENT} & |\widehat{H}_\ell| > \varepsilon^*(\ell). \end{cases} \tag{11}$$

UNDECIDABLE denotes that the data supplied do not determine the classification; it is not used in the sense of computability theory.

**Theorem 2.27** (Properties of the three-valued classification)**.**

*(i) The three cases of (11) are exhaustive and mutually exclusive.*

*(ii) The class UNDECIDABLE is non-empty whenever $\varepsilon_{\mathrm{data}}(\ell) > 0$, that is, for every cycle whose reaction data carry independently estimated uncertainty. In realistic cases $\varepsilon_{\mathrm{data}}$ exceeds $\varepsilon_{\mathrm{num}}$ by many orders of magnitude.*

*(iii) Any fixed rule that reduces the three classes to two either classifies as consistent some cycles whose sum exceeds the rounding-error bound, or classifies as inconsistent some cycles whose sum lies within the data uncertainty. Neither classification is supported by the data.*

*Proof.* See Supplementary Material, Section S10.

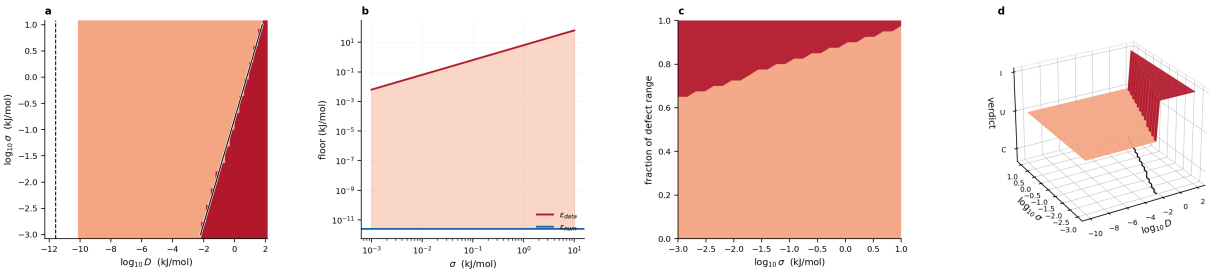



Figure 2: **Three-valued classification as a function of data uncertainty and defect magnitude.** A six-edge cycle at $\Lambda = 200$ kJ/mol on a $26 \times 40$ grid of per-reaction uncertainty $\sigma \in [10^{-3}, 10]$ against injected defect $D \in [10^{-10}, 10^{2}]$ kJ/mol (1040 points), with noise-free reaction data, so that each verdict is determined by $D$ and the two components of the tolerance. (**A**) Verdict map: blue CONSISTENT, orange UNDECIDABLE, red INCONSISTENT. The solid curve is $\varepsilon_{\text{data}}(\sigma)$, which equals $\varepsilon^*$ to within $10^{-9}$; the dashed vertical line is $\varepsilon_{\text{num}} = 2.48 \times 10^{-12}$ kJ/mol. The orange region between them covers 80.9% of the grid. (**B**) The two components against $\sigma$. The rounding-error bound is constant at $2.48 \times 10^{-12}$; the uncertainty threshold increases from $6.31 \times 10^{-3}$ to 63.1 kJ/mol. At the midpoint of the range they differ by 11.5 orders of magnitude (Theorem 2.27(ii)). (**C**) Proportion of each verdict against $\sigma$. The INCONSISTENT proportion decreases from 0.350 to 0.025 as data uncertainty increases; at $\sigma = 10$ the test returns UNDECIDABLE on 97.5% of the defect range, all of which a two-valued rule would have classified. (**D**) The verdict as a surface over $(\log_{10} D, \log_{10} \sigma)$. No CONSISTENT region appears because the sweep begins at $D = 10^{-10}$, forty times $\varepsilon_{\text{num}}$ (Section 3.2).

*Remark* 2.28 (Information carried by the intermediate class). The class UNDECIDABLE conveys information that a two-valued test discards: the cycle sum is too large to be rounding error and too small to be resolved by data of the given quality. It also specifies what would resolve it, namely a reduction of the uncertainties $\sigma_e$ on the reactions of $\ell$ to below $|\widehat{H}_\ell|/\varepsilon_{\text{data}}(\ell)$ of their present values.

## 2.5 Dependence on the choice of cycle basis

### 2.5.1 Invariance of the network-level classification

**Theorem 2.29** (Basis independence of the verdict)**.** *Let $\mathcal{B}_1, \mathcal{B}_2$ be cycle bases of $G$ and let $d$ be reaction data. Then $z^\top d = 0$ for all $z \in \mathcal{B}_1$ if and only if $z^\top d = 0$ for all $z \in \mathcal{B}_2$. Consequently the classification of the network as consistent or inconsistent does not depend on the basis.*

*Proof.* See Supplementary Material, Section S11.

### 2.5.2 Basis dependence of the flagged cycles

**Proposition 2.30** (Basis dependence of the flagged set)**.** *There exist a graph $G$, reaction data $d$ with a single defective edge $e^*$, and cycle bases $\mathcal{B}_1, \mathcal{B}_2$ of $G$ such that exactly one element of $\mathcal{B}_1$ has a nonzero cycle sum, while three elements of $\mathcal{B}_2$ do.*

*Proof.* See Supplementary Material, Section S12.

*Remark* 2.31 (Consequence for the reporting of localisation). By Proposition 2.30, the statement that a defect is localised to cycle $\ell$ depends both on the network and on the basis computed by the implementation. A protocol that reports localisation must therefore either fix a canonical basis and state it, or report a basis-independent quantity. The second approach is adopted here.

### 2.5.3 Witness set

**Definition 2.32** (Flagged set and witness set)**.** Fix a cycle basis $\mathcal{B}$ and let $F(\mathcal{B}) = \{\ell \in \mathcal{B} : \text{verdict}(\ell) = \mathsf{INCONSISTENT}\}$ be the flagged set. The *witness set* is

$$W(\mathcal{B}) \;=\; \bigcap_{\ell \in F(\mathcal{B})} E(\ell), \tag{12}$$

the set of edges that lie on every flagged cycle, with $W = \emptyset$ when $F = \emptyset$.

**Theorem 2.33** (Containment of the defective edge in the witness set)**.** *Suppose that exactly one edge $e^*$ is defective, with defect $D > 2\max_\ell \varepsilon^*(\ell)$, and that the remaining reaction data are exact. Then for every cycle basis $\mathcal{B}$, $e^* \in W(\mathcal{B})$, and $F(\mathcal{B})$ is the set of basis cycles with a nonzero net traversal of $e^*$; for simple cycles, these are the cycles containing $e^*$. If the reaction data carry independent errors as in Theorem 2.19, the same holds with probability at least $1 - \nu\alpha$.*

*Proof.* See Supplementary Material, Section S13.

**Corollary 2.34** (Basis-independent localisation)**.** *The basis-independent content of a positive result is that the defect lies in $W(\mathcal{B})$. The cardinality $|W|$ measures the resolution of the localisation: $|W| = 1$ identifies a single reaction, and a large $|W|$ indicates that the flagged cycles share many edges, which the test cannot distinguish.*

*Remark* 2.35 (Effect of a minimum cycle basis). Theorem 2.33 holds for every basis, but short cycles share fewer edges, so a minimum-weight cycle basis (a basis of least total length; Horton, 1987; Kavitha et al., 2009) generally yields a smaller $W$ and hence a finer localisation. Such a basis is used in the protocol of Section 2.8, and $|W|$ is reported as a primary outcome in place of a single cycle.

## 2.6 Validity of the reference network

The cycle-consistency test presupposes that the network as supplied admits a potential function. If the thermodynamic annotations of the reconstruction are themselves inconsistent, cycles are flagged for reasons unrelated to any disease state, and the test provides no diagnostic information.

**Definition 2.36** (Validity check)**.** A network with reaction data $d$ *passes the validity check* if no cycle $\ell$ of the chosen basis receives the verdict INCONSISTENT on the reference (nominally healthy) parameter set, with each $\varepsilon_{\text{data}}(\ell)$ computed at level $\alpha/\nu$, so that a consistent reference network passes with probability at least $1 - \alpha$.

The check requires the absence of INCONSISTENT verdicts rather than CONSISTENT verdicts throughout, because a consistent network with uncertain reaction data is expected to return UNDECIDABLE; CONSISTENT is attainable only for data that are exactly potential differences.

**Criterion 2.37** (Invalid versus negative)**.** An experiment in which the reference network fails Definition 2.36 is **invalid**, not negative. It shows that the thermodynamic annotations of the reconstruction are not consistent to the precision they report, and it supports no conclusion about the diagnostic. An experiment in which the reference network passes and a perturbed network is not flagged is **negative**: it shows that the test does not detect that perturbation at that data quality.

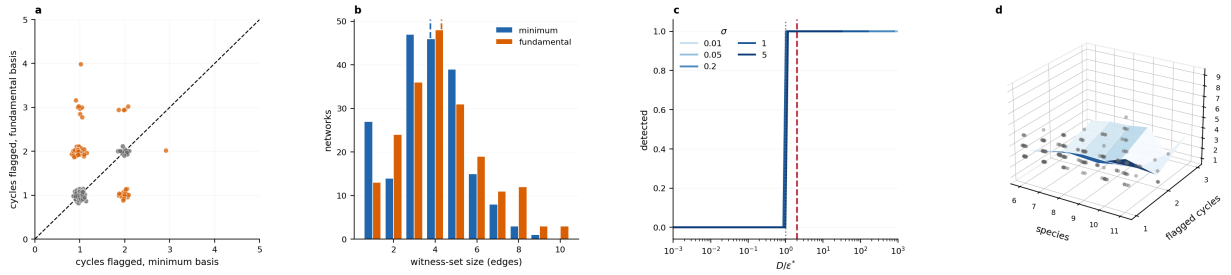



Figure 3: **Basis dependence of localisation and invariance of the witness set.** Panels (A), (B) and (D) use 200 random networks of 6 to 11 species with 2 to 4 chords (edges outside a spanning tree), each with one edge defect of 50 kJ/mol and $\sigma = 0.05$ kJ/mol; panel (C) is an independent detection sweep. (**A**) Number of cycles flagged under a minimum-weight basis against a fundamental basis (the basis formed by one spanning tree and its chords), with jitter to separate coincident integer points. Orange points indicate disagreement: 69/200 trials (34.5%). The network-level verdict agreed in 200/200 trials (Theorem 2.29). (**B**) Witness-set cardinality $|W| = |\bigcap_{\ell \in F} E(\ell)|$ for each basis, with means shown dashed. The minimum-weight basis gives a finer localisation (3.77 against 4.30 edges; Remark 2.35), by less than half an edge. Both bases contained the defective edge in 200/200 trials (Theorem 2.33). (**C**) Detection rate against $D/\varepsilon^*$ for five data qualities $\sigma \in \{0.01, 0.05, 0.2, 1, 5\}$ kJ/mol, 220 points each, on noise-free reaction data. The dotted vertical line marks $\varepsilon^*$ and the dashed line the detection limit $2\varepsilon^*$ of Theorem 2.24. Without noise the verdict depends on $D$ only through $D/\varepsilon^*$, so the curves coincide by construction and change at $\varepsilon^*$ (0.51 to 0.58 of $2\varepsilon^*$, the spread reflecting the spacing of the $D$ grid); the guarantee under noise is tested in V6 (Section 3.2). (**D**) Mean witness-set size against network size and number of flagged cycles. Localisation becomes coarser as more cycles are flagged (Pearson $r = -0.72$): because the witness set is an intersection, fine localisation requires short cycles with little overlap.

*Remark* 2.38 (Pre-registration of the criterion). If Criterion 2.37 is not stated in advance, a failed reference check can be attributed to the data when the diagnostic succeeds and to the method when it fails. Fixing the interpretation before the data are examined removes this freedom of interpretation (Sandve, Nekrutenko, Taylor, & Hovig, 2013; Simmons, Nelson, & Simonsohn, 2011).

## 2.7 Experimental data

**Cycles in a stoichiometric network.** Most reactions have several substrates or products, so a realistic network is described by a stoichiometric matrix $S$ (species by reactions) rather than by a graph; Section 2.1 treats the special case $S = B$. A cycle is an integer vector $z \in \ker S$, and its sum is $H_z = z^{\top} d$. All results above carry over with integer coefficients $z_{\ell}(e)$ in place of $\pm 1$: if $d = S^{\top}\mu$ then $H_z = (Sz)^{\top}\mu = 0$, which gives Lemmas 2.8 and 2.17, and the proof of Theorem 2.19 is unchanged. For reaction data supplied directly, the rounding-error bound is $\varepsilon_{\mathrm{num}}(z) = \gamma_{|z|} \sum_e |z_e d_e|$, with $|z|$ the number of reactions in the cycle (Supplementary Section S17). The kernel of $S$ is spanned by an integer basis computed in exact rational arithmetic.

**Equilibrium constants from TECRDB.** The Thermodynamics of Enzyme-Catalyzed Reactions Database (TECRDB; R. N. Goldberg, Tewari, & Bhat, 2004) records apparent equilibrium constants $K'$ measured for individual reactions, each with the pH, ionic strength, temperature and pMg (the negative decadic logarithm of the free magnesium concentration) of the measurement. Measurements graded A to C at 293 to 313 K were admitted. Each reaction was balanced with water and protons, and $-RT \ln K'$ was converted by the reverse Legendre transform to the standard reaction Gibbs energy in the chemical reference state at the conditions of measurement, following the procedure by which the component-contribution method prepares its training data (Beber et al., 2022; Noor et al., 2013). The resulting values $d_e = \Delta_r G^{\circ}_e$ are independent measurements and not differences of formation energies, which would cancel on every cycle and make the test vacuous. Each study contributes one value per reaction, and $\sigma_e = \sigma/\sqrt{n_e}$ for a reaction measured in $n_e$ studies, with $\sigma$ the pooled between-study standard deviation.

**Human erythrocyte reaction network.** For 38 enzymes covering the published erythrocyte models (Mulquiney & Kuchel, 1999), every catalytic activity annotated in UniProtKB/Swiss-Prot (The UniProt Consortium, 2025) was retrieved together with the stoichiometry of its Rhea reaction (Bansal et al., 2022). Compounds were identified by molecular connectivity, so that anomers (stereoisomers of a sugar differing only at the anomeric carbon) are merged, and ion pumps were reduced to their net chemical reaction. Reactions with generic participants were excluded. All data were retrieved on 21 October 2026; the snapshot, its checksums and the analysis are provided in the repository.

## 2.8 Implementation and protocol

**Typed implementation.** A correctly computed $\varepsilon^{*}$ is of no use if the comparison is subsequently written against a literal constant, as in `if abs(H) > 1e-6`. The test is therefore expressed in a small typed language in which the only operation producing a `Verdict` requires a `Tolerance` as an argument. Consequently, no well-typed program yields a verdict without its tolerance, and the comparison of a cycle sum with a numeric literal is a type error (Supplementary Section S15). The type system guarantees that the tolerance is computed and recorded; it does not guarantee that the inputs are correct, and a program supplied with an incorrect $\sigma_e$ is well-typed and gives an incorrect result.

**Experimental protocol and pre-registered criteria.** Supplementary Section S16 specifies the protocol in executable detail: the network (Brunk et al., 2018); the thermodynamic (Flamholz

Table 1: Measured rounding error of cycle sums of consistent circuits, whose exact value is zero, so that the computed value equals the error. Binary64. Each entry is the maximum over 60 random circuits with potentials drawn uniformly from $[-\Lambda, \Lambda]$, so that $\Lambda$ also determines their spread. The error increases with both $L$ and $\Lambda$, as Theorem 2.12 predicts, over three orders of magnitude; its largest value is $2.7 \times 10^{-12}$.

| | $\Lambda$ (kJ/mol) | | | |
|---|---|---|---|---|
| $L$ | 10 | 100 | 1000 | 3000 |
| 3 | $3.6 \times 10^{-15}$ | $2.8 \times 10^{-14}$ | $2.3 \times 10^{-13}$ | $9.1 \times 10^{-13}$ |
| 8 | $3.6 \times 10^{-15}$ | $2.8 \times 10^{-14}$ | $3.4 \times 10^{-13}$ | $1.8 \times 10^{-12}$ |
| 16 | $5.3 \times 10^{-15}$ | $5.7 \times 10^{-14}$ | $6.8 \times 10^{-13}$ | $1.8 \times 10^{-12}$ |
| 30 | $5.3 \times 10^{-15}$ | $5.7 \times 10^{-14}$ | $6.8 \times 10^{-13}$ | $2.7 \times 10^{-12}$ |

et al., 2012; Noor et al., 2013) and physiological (Mulquiney & Kuchel, 1999; Rapoport & Heinrich, 1975) data; and a perturbation of the forward rate constant of pyruvate kinase alone, which violates the Haldane relation and introduces a defect of $RT \ln(1/0.15) \approx 4.9$ kJ/mol on that edge. Three controls are included, each designed so that it can fail on correctly formed input, and the criteria for success and failure are fixed in advance. One control scales the forward and reverse rate constants by the same factor, corresponding to the loss of activity observed in clinical pyruvate kinase deficiency (Zanella, Fermo, Bianchi, & Valentini, 2005); this perturbation preserves every cycle sum and must not be flagged. The criteria distinguish a *negative* experiment from an *invalid* one: by Criterion 2.37, a reference network that fails its consistency check supports no conclusion about the diagnostic. Applied to the erythrocyte network of Section 2.7, the protocol terminates at its third step, which requires the perturbed edge to lie on a cycle (Section 3.3).

# 3 Results and Discussion

Section 3.1 quantifies the performance of a fixed tolerance. Section 3.2 reports the computational validation of the numbered results, Section 3.3 applies the test to experimental data, and Section 3.4 discusses the limitations of the approach.

## 3.1 Performance of a fixed tolerance

All numbers in this section are produced by the validation suite described in Section 3.2, and the construction is specified in full so that it can be reproduced exactly.

### 3.1.1 Magnitude of rounding error in double precision

Table 1 reports the measured rounding error. In 1920 trials covering $L \in [3, 30]$ and $\Lambda \in [10, 3000]$ kJ/mol, the bound of Theorem 2.12 was never violated. The bound overestimates the error: its median ratio to the observed error is 729, and the ratio increases with $L$, from about 180 at $L = 3$ to 2200 at $L = 30$ for $\Lambda = 1000$ kJ/mol, because the bound grows as $L^2$ while random rounding errors accumulate more slowly (Higham, 2002).

The largest rounding error in Table 1, $2.7 \times 10^{-12}$, lies six orders of magnitude below $10^{-6}$. In binary64 a fixed threshold of $10^{-6}$ therefore cannot flag any consistent circuit of the grid as a result of rounding, and the measured false-positive rate was 0.000 in all four strata of Figure 4A (1600 trials).

*Remark* 3.1 (Position of the conventional threshold). Corollary 2.13 holds: the rounding error varies with $L\Lambda$ over three orders of magnitude in Table 1. The corollary does not imply that the conventional threshold lies within this range, and in binary64 it lies above all of it. The case for a computed tolerance in double precision therefore rests on the uncertainty of the data and on detection sensitivity, not on rounding.

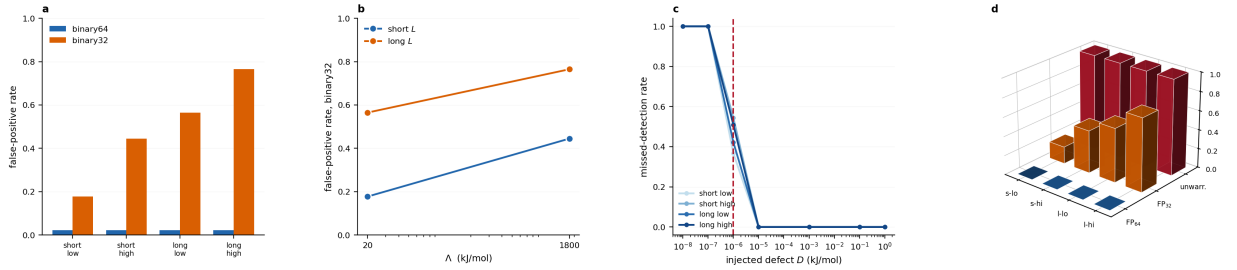



Figure 4: **Performance of a fixed tolerance $\varepsilon = 10^{-6}$.** Four strata of 400 trials, combining short ($L \in [3, 5]$) or long ($L \in [15, 22]$) cycles with a low ($\Lambda = 20$ kJ/mol) or high ($\Lambda = 1800$ kJ/mol) potential range. (**A**) False-positive rate on consistent circuits: 0.000 in every stratum in binary64 (bars of zero height; Remark 3.1); the same null experiment in binary32 gives 0.177, 0.445, 0.565 and 0.765. (**B**) The binary32 rate by length class against $\Lambda$. The length has the larger effect, so the rate is not monotone in $L\Lambda$. (**C**) Missed-detection rate against injected defect $D \in [10^{-8}, 1]$ kJ/mol, with $\varepsilon$ dashed. Every defect below the threshold is missed in every stratum, and none is missed at $D = 10^{-5}$. The threshold acts as a detection limit rather than as a filter for rounding error. (**D**) The three error types per stratum. With $\sigma = 1$ kJ/mol and a defect of 0.5 kJ/mol, which lies within that uncertainty, the fixed test returns INCONSISTENT on every cycle for which the cycle-specific test returns UNDECIDABLE. This, rather than the false-positive rate, is the principal deficiency of the fixed tolerance.

### 3.1.2 Missed detections and unsupported positive results

Figure 4 distinguishes the errors that a fixed tolerance makes from those it does not make.

**Detection limit imposed by the threshold.** At $D = 10^{-7}$ the missed-detection rate is 1.000 in every stratum (Figure 4C), although these cycle sums are computed almost exactly. The misses result from a threshold chosen by convention rather than from the smallest defect of interest. A numerical solver setting thus imposes a detection limit of $10^{-6}$ kJ/mol that has no biochemical justification.

**Classification of defects within the data uncertainty.** With a per-reaction uncertainty of $\sigma = 1$ kJ/mol and an injected defect of 0.5 kJ/mol, the fixed test returns INCONSISTENT on 100% of cycles, since $0.5 \gg 10^{-6}$, whereas the cycle-specific test returns UNDECIDABLE (Figure 4D). A two-valued test cannot express that a cycle sum lies far above the rounding error and far below the resolution of the data, and is forced to report a result that the data do not support. Section 3.3 shows the same behaviour on measured data.

**False positives in single precision.** When the null experiment is repeated in binary32, where $u \approx 6 \times 10^{-8}$ and the rounding-error bound approaches $10^{-6}$, the false-positive rate increases from 0.177 to 0.765 with both $L$ and $\Lambda$, and is not monotone in their product (Figure 4B), since $L$ enters Theorem 2.12 quadratically and $\Lambda$ linearly. $\varepsilon_{\text{num}}$ must therefore be computed for each cycle from both quantities.

## 3.2 Computational validation

Each numbered result of Section 2 was checked computationally. Table 2 summarises the experiments.

Table 2: Validation suite. Every experiment is seeded, and the JSON record is the source of every number reported in this article.

| | Result tested | Trials | Outcome |
|---|---|---|---|
| V1 | Theorem 2.12 | 1920 | 0 violations; median ratio 729 |
| V2 | Section 3.1 | 1600 | no false positives in binary64 |
| V3 | Theorem 2.27 | 1040 | noise-free map; UNDECIDABLE = 80.9% |
| V4 | Theorem 2.29 and proposition 2.30 | 200 | verdicts agree; flagged sets differ in 34.5% |
| V5 | Theorem 2.24 | 1100 | noise-free; 0 violations |
| V6 | Lemma 2.17, Theorems 2.19, 2.23 and 2.24 | 24,000 | FP 0.009 at $\alpha = 0.01$; detection $\geq 0.993$ |

### 3.2.1 Design of the validation suite

For an identity, a computational check verifies the implementation; for a bound, it measures the overestimation factor (V1). Experiments V3 and V5 use noise-free reaction data, for which each verdict is a deterministic function of the defect and the two components of the tolerance; they therefore map the structure of the classification rather than test a probabilistic claim. The probabilistic statements are tested by simulating noise in V6. Defects are always introduced on an edge (Remark 3.2), and V1, V4 and V6 each contain a negative control, a check that an effect does not occur on correctly formed input.

*Remark* 3.2 (Introduction of defects on reactions). A defect cannot be introduced by shifting a species potential $\mu_i$: the shifted reaction data remain potential differences, so by Lemma 2.8 every cycle sum remains zero and no defect is detected. An inconsistency can be introduced only on an edge (Definition 2.9). Because a perturbation of a species potential produces a null result indistinguishable from a genuine negative, V6 contains a negative control verifying that such a perturbation is not detected.

### 3.2.2 Extent of the undecidable region (V3)

Theorem 2.27(ii) states only that the class UNDECIDABLE is non-empty. In a sweep of $\sigma \in [10^{-3}, 10]$ kJ/mol against $D \in [10^{-10}, 10^{2}]$ kJ/mol on a six-membered cycle at $\Lambda = 200$ kJ/mol (1040 grid points; Figure 2), the verdict was UNDECIDABLE on 80.9% of the grid and INCONSISTENT on 19.1%. These proportions depend on the limits of the sweep; the separation between the two components of the tolerance, 11.5 orders of magnitude near the middle of the $\sigma$ range, does not. The CONSISTENT proportion of 0.000 results from the lower limit of the sweep, which is forty times the rounding-error bound.

### 3.2.3 Basis dependence of localisation (V4)

On 200 random networks of 6 to 11 species with 2 to 4 chords, each carrying a single edge defect of 50 kJ/mol (Figure 3A,B), the network-level verdict agreed between a minimum-weight and a fundamental basis in all trials (0/200 disagreements), as Theorem 2.29 requires. The two bases flagged different numbers of cycles in 69/200 trials (34.5%), so a report localising the defect to a single cycle would have depended on the basis in about one trial in three (Proposition 2.30). The witness set contained the defective edge in 200/200 trials under both bases (Theorem 2.33). The minimum-weight basis gave a finer localisation, as Remark 2.35 predicts, by less than half an edge (3.77 against 4.30 edges). A witness set of about four edges identifies a region of the network rather than a single reaction, and is reported as such.

### 3.2.4 Detection with and without data noise (V5, V6)

Theorem 2.24 guarantees detection when $D > 2\varepsilon^{*}$. On noise-free reaction data (V5), with $D$ swept over 220 points at each of five data qualities, $\sigma \in \{0.01, 0.05, 0.2, 1, 5\}$ kJ/mol, the guarantee was

violated in none of 1100 evaluations (Figure 3C). Without noise a defect is detected as soon as it exceeds $\varepsilon^*$, that is, at 0.51 to 0.58 of $2\varepsilon^*$, the spread reflecting the grid spacing. The factor of two is required only for the guarantee under noise (Remark 2.25).

V6 introduces noise on ring cycles of length $L \in \{4, 8, 16\}$ with $\sigma \in \{0.1, 1\}$ kJ/mol, 4000 trials per configuration and $\alpha = 0.01$. Independent normal errors on the species potentials left the verdict CONSISTENT in all 24,000 trials, as Lemma 2.17 requires, whereas a model of uncertainty attached to the species would predict a cycle-sum standard deviation of $\sqrt{L}\,\sigma$. Independent normal errors on the reaction data gave false-positive rates of 0.008 to 0.011 (mean 0.009), against the nominal 0.01 of Theorem 2.23, and a defect of $2\varepsilon^*$ was detected with frequency 0.9935 to 0.996, against the guaranteed 0.99 of Theorem 2.24. A defect of $\varepsilon^*$, below the detection limit, was detected with frequency 0.49 to 0.51.

### 3.3 Application to measured data

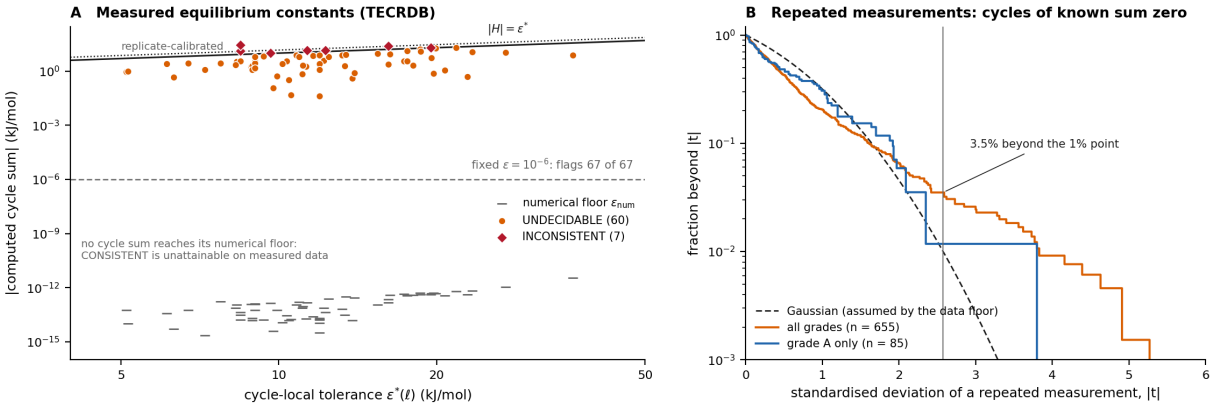



Figure 5: **The test applied to measured equilibrium constants.** (**A**) Each of the 67 cycles spanning the cycle space of the 418 reactions measured in TECRDB: computed sum against cycle-specific tolerance (Gaussian uncertainty threshold), with the rounding-error bound (grey dash) fourteen orders of magnitude lower. Every sum exceeds the fixed $\varepsilon = 10^{-6}$; 7 exceed $\varepsilon^*$ (solid line) and 3 exceed the tolerance calibrated on replicate measurements (dotted line). (**B**) Leave-one-out deviations between studies of the same reaction, which form cycles of known sum zero. Across all quality grades the tail is heavier than the Gaussian distribution assumed by the uncertainty threshold; grade-A data follow the Gaussian distribution.

**Fixed tolerances on TECRDB cycles.** The admitted TECRDB data comprise 3,276 measurements from 908 studies of 418 reactions, of which 165 were measured in more than one study (between-study standard deviation $\sigma = 2.68$ kJ/mol). The cycle space has dimension 67, with cycles of 3 to 25 reactions (Figure 5A). All cycle sums lie between 0.042 and 27.8 kJ/mol. A fixed tolerance of $10^{-6}$ or $10^{-3}$ therefore classifies all 67 cycles as inconsistent, and tolerances of 1, 5 or 10 kJ/mol classify 54, 27 or 14 as inconsistent: the outcome is determined by the choice of $\varepsilon$. The rounding-error bounds lie between $2.3 \times 10^{-15}$ and $3.4 \times 10^{-12}$ kJ/mol and the uncertainty thresholds between 5.1 and 36.4 kJ/mol, a median separation of 14.1 orders of magnitude, so that CONSISTENT cannot be attained on measured data. The cycle-specific tolerance returns UNDECIDABLE for 60 cycles and INCONSISTENT for 7.

**Calibration of the uncertainty threshold.** Theorem 2.19 uses a Gaussian quantile. Two studies of the same reaction form a cycle whose true sum is zero, so the 655 leave-one-out deviations of a study from the remaining studies test this assumption on the data themselves (Figure 5B). They exceed the Gaussian 1% point in 3.5% of cases, and their 99th percentile is 3.81

standard deviations instead of 2.58. With this empirical quantile the test flags 3 cycles; with the distribution-free quantile $\alpha^{-1/2} = 10$ given by Chebyshev's inequality (Supplementary Section S7) it flags none. The heavy tail originates in the lower quality grades: restricted to grade A, the replicates follow the Gaussian distribution (1.2% beyond its 1% point, $\sigma = 1.49$ kJ/mol), and 2 of 13 cycles are flagged under either quantile. V6 establishes the false-positive rate under Gaussian noise only; for measured data the quantile must first be calibrated on replicate measurements.

**Localisation with multiple defects.** Over 200 fundamental bases of the same cycle space, at least one cycle was flagged in every basis, as Theorem 2.29 requires, but the number of flagged cycles ranged from 2 to 40 and the witness set of Definition 2.32 was empty. The data therefore contain several independent inconsistencies, a case outside the scope of Theorem 2.33 (Section 3.4). In this case the quantity that localises is the frequency with which a reaction lies on a flagged cycle. The single measurement of pyrophosphate–fructose-6-phosphate 1-phosphotransferase lies on a flagged cycle in 94% of bases; its cycle with inorganic pyrophosphatase and fructose-bisphosphatase sums to $-27.8$ kJ/mol, 8.5 standard deviations from zero (Supplementary Section S17).

**Human erythrocyte network.** The 38 enzymes carry 118 annotated reactions, of which 70 are chemically distinct and fully specified, over 101 species. The cycle space has dimension 7, and only 19 of the 70 reactions lie on a cycle. A reaction that lies on no cycle does not enter any cycle sum, so no cycle-consistency test can detect a defect on it, regardless of the data. Pyruvate kinase is such a reaction, since no erythrocyte enzyme converts pyruvate back to phosphoenolpyruvate. The same holds for every glycolytic step except phosphoglycerate kinase and phosphoglycerate mutase, which lie on the only cycle through glycolysis: the Rapoport–Luebering shunt, the bypass of phosphoglycerate kinase through 2,3-bisphosphoglycerate, closed by ATP hydrolysis. TECRDB contains no measurement for bisphosphoglycerate mutase or bisphosphoglycerate phosphatase, the enzymes of the shunt; the other six cycles pass through nucleotide kinases, and none of the seven cycles has a measured equilibrium constant for every reaction. The protocol accordingly terminates at its third step, as pre-registered. With the present data the test cannot be applied to the erythrocyte; measurements for the two enzymes of the shunt would make one cycle testable.

### 3.4 Limitations

**Conditions of measurement.** A correction applied to species potentials, such as an activity coefficient, cancels around every cycle (Lemma 2.17). Differences between the conditions under which individual reaction data were obtained do not cancel. Section 2.7 removes the dependence on pH, ionic strength and magnesium by the reverse Legendre transform, but not the temperature dependence of $\Delta_r G^\circ$, which is limited by admitting only measurements at 293 to 313 K. Where TECRDB reports no ionic strength or magnesium concentration, $I = 0.25$ M and the absence of magnesium are assumed. Both residual effects are systematic, contribute to the between-study spread, and are absorbed by the calibration described in Section 3.3.

**Steady-state assumption.** The circuit representation of Proposition 2.4 holds at steady state. A network under transient conditions has no time-invariant potential function, and cycle sums computed from instantaneous concentrations are not the quantity analysed here.

**Correlated uncertainties.** Theorem 2.19 assumes that the errors of the reaction data are independent. Two measurements from the same laboratory are not independent, and each study therefore contributes one value per reaction (Section 2.7). Reaction Gibbs energies estimated from shared group contributions are also correlated (Noor et al., 2013), but the part of the correlation that arises from common formation energies cancels around every cycle (Lemma 2.17).

**Defects undetectable by cycle sums.** The test detects reaction data that are not potential differences, on reactions that lie on a cycle, and no other perturbation. A loss of enzyme activity that scales the forward and reverse rate constants by the same factor, which is the usual molecular consequence of an enzyme deficiency, preserves the Haldane relation and therefore every cycle sum; so does any perturbation of concentrations alone (Lemma 2.17). A reaction that lies on no cycle, such as pyruvate kinase in the erythrocyte, cannot be tested at all.

**Network assembly.** The erythrocyte network merges anomers and reduces ion pumps to their net chemical reaction, neglecting the membrane potential. Both simplifications can only add cycles, so they do not affect the finding that pyruvate kinase lies on no cycle.

**Multiple defects.** Theorem 2.33 assumes a single defective edge. Measured data contain several, and the witness set is then empty (Section 3.3). The appropriate generalisation is the family of minimal edge sets that intersect every flagged cycle, which is a hitting-set problem and is not developed here; the frequency with which an edge lies on a flagged cycle serves as a heuristic substitute.

**Conservatism of the rounding-error bound.** Equation (7) is a standard worst-case bound, quadratic in $L$, and exceeds the observed error by one to three orders of magnitude (V1). Since $\varepsilon_{\text{data}}$ exceeds it by fourteen orders of magnitude on measured data, this conservatism changes no verdict reported here.

# 4 Conclusion

The cycle sum of a thermodynamically consistent metabolic network is zero as an algebraic identity. A diagnostic based on it must therefore distinguish rounding error and data uncertainty from a genuine defect, and a fixed tolerance treated as a solver setting does not make this distinction. In double precision the conventional $\varepsilon = 10^{-6}$ produces no false positives from rounding; its deficiency is that it classifies as inconsistent cycle sums that the data cannot resolve, and on measured equilibrium constants all 67 cycle sums exceed it. The cycle-specific tolerance replaces it by the sum of a rounding-error bound and an uncertainty threshold derived from reaction-level uncertainty alone. The separation of fourteen orders of magnitude between the two components makes the classification three-valued: on measured data, 60 cycles are undecidable and 7 inconsistent.

The analysis of experimental data required two refinements of the method. The Gaussian quantile of the uncertainty threshold is too small for the database as a whole, and replicate measurements, which form cycles of known sum zero, provide the calibration. The witness set assumes a single defect, whereas the measured data contain several. The experimental data also delimit the application. In the human erythrocyte, pyruvate kinase lies on no cycle and no cycle is fully measured, so no erythrocyte defect can be examined by a cycle-consistency test with the present data; measurements for the two enzymes of the Rapoport–Luebering shunt would make one cycle testable. Whether disease states produce reaction data that are not potential differences remains an open question.

# Reproducibility

The validation suite of Section 3.2 is in `validation/run_validation.py`, is seeded (`seed` = 20260819), and writes every quantity reported in this article to `validation/results.json`; the tables of Sections 3.1 and 3.2 are read from that file. Figures 1–4 and the graphical abstract are generated from it (the latter also from `results_real.json`) by `validation/make_panels.py` and `validation/make_graphical_abstract.py`, so no plotted value is computed independently of a reported one. The analysis of Section 3.3 is in `validation/realdata/`: `fetch_sources.py`

saves the TECRDB, UniProt and Rhea data with their SHA-256 checksums, `run_realdata.py` writes `results_real.json`, and Figure 5 is drawn from it. Its first stage needs the eQuilibrator compound cache (Zenodo, doi:10.5281/zenodo.4128543); the second runs from the derived tables it writes. No step requires proprietary software, wet-lab work, or computation beyond a single workstation.

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## Ethical Statement

This study does not contain any studies with human or animal subjects performed by the author. All data used are computationally generated or drawn from public databases of previously published measurements.

## Conflicts of Interest

The author declares no conflict of interest.

## Data Availability Statement

The data supporting the findings of this study are openly available. The seeded validation suite (`run_validation.py`), the complete record of every quantity reported here (`results.json`), and the figure-generation scripts are deposited in a public repository, together with the snapshot of experimental data analysed in Section 3.3, retrieved on 21 October 2026 from TECRDB (R. N. Goldberg et al., 2004), UniProtKB (The UniProt Consortium, 2025) and Rhea (Bansal et al., 2022), all publicly available, and the tables derived from it.

## Author Contribution Statement

**Kundai Farai Sachikonye:** Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Visualization, Writing – original draft, Writing – review & editing. The author has read and agreed to the published version of the manuscript.

## References

Alberty, R. A. (1998). Calculation of standard transformed Gibbs energies and standard transformed enthalpies of biochemical reactants. *Archives of Biochemistry and Biophysics*, *353*(1), 116–130. doi: 10.1006/abbi.1998.0638

Atkins, P., de Paula, J., & Keeler, J. (2018). *Atkins' physical chemistry* (11th ed.). Oxford: Oxford University Press.

Bansal, P., Morgat, A., Axelsen, K. B., Muthukrishnan, V., Coudert, E., Aimo, L., ... Bridge, A. (2022). Rhea, the reaction knowledgebase in 2022. *Nucleic Acids Research*, *50*(D1), D693–D700. doi: 10.1093/nar/gkab1016

Beard, D. A., Liang, S.-d., & Qian, H. (2002). Energy balance for analysis of complex metabolic networks. *Biophysical Journal*, *83*(1), 79–86. doi: 10.1016/S0006-3495(02)75150-3

Beber, M. E., Gollub, M. G., Mozaffari, D., Shebek, K. M., Flamholz, A. I., Milo, R., & Noor, E. (2022). eQuilibrator 3.0: a database solution for thermodynamic constant estimation. *Nucleic Acids Research*, *50*(D1), D603–D609. doi: 10.1093/nar/gkab1106

Bollobás, B. (1998). *Modern graph theory* (Vol. 184). Springer.

Brunk, E., Sahoo, S., Zielinski, D. C., Altunkaya, A., Dräger, A., Mih, N., ... Palsson, B. O. (2018). Recon3D enables a three-dimensional view of gene variation in human metabolism. *Nature Biotechnology*, *36*(3), 272–281. doi: 10.1038/nbt.4072

Currie, L. A. (1968). Limits for qualitative detection and quantitative determination. Application to radiochemistry. *Analytical Chemistry*, *40*(3), 586–593. doi: 10.1021/ac60259a007

Diestel, R. (2017). *Graph theory* (5th ed., Vol. 173). Springer.

Flamholz, A., Noor, E., Bar-Even, A., & Milo, R. (2012). eQuilibrator — the biochemical thermodynamics calculator. *Nucleic Acids Research*, *40*(D1), D770–D775. doi: 10.1093/nar/gkr874

Goldberg, D. (1991). What every computer scientist should know about floating-point arithmetic. *ACM Computing Surveys*, *23*(1), 5–48. doi: 10.1145/103162.103163

Goldberg, R. N., Tewari, Y. B., & Bhat, T. N. (2004). Thermodynamics of enzyme-catalyzed reactions—a database for quantitative biochemistry. *Bioinformatics*, *20*(16), 2874–2877. doi: 10.1093/bioinformatics/bth314

Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th python in science conference* (pp. 11–15).

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*, 357–362. doi: 10.1038/s41586-020-2649-2

Higham, N. J. (2002). *Accuracy and stability of numerical algorithms* (2nd ed.). Philadelphia: SIAM. doi: 10.1137/1.9780898718027

Horton, J. D. (1987). A polynomial-time algorithm to find the shortest cycle basis of a graph. *SIAM Journal on Computing*, *16*(2), 358–366. doi: 10.1137/0216026

IEEE. (2019). *IEEE standard for floating-point arithmetic, IEEE std 754-2019*. New York: IEEE. doi: 10.1109/IEEESTD.2019.8766229

Kavitha, T., Liebchen, C., Mehlhorn, K., Michail, D., Rizzi, R., Ueckerdt, T., & Zweig, K. A. (2009). Cycle bases in graphs: characterization, algorithms, complexity, and applications. *Computer Science Review*, *3*(4), 199–243. doi: 10.1016/j.cosrev.2009.08.001

Mulquiney, P. J., & Kuchel, P. W. (1999). Model of 2,3-bisphosphoglycerate metabolism in the human erythrocyte based on detailed enzyme kinetic equations. *Biochemical Journal*, *342*(3), 581–596. doi: 10.1042/bj3420581

Noor, E., Haraldsdóttir, H. S., Milo, R., & Fleming, R. M. T. (2013). Consistent estimation of Gibbs energy using component contributions. *PLoS Computational Biology*, *9*(7), e1003098. doi: 10.1371/journal.pcbi.1003098

Orth, J. D., Thiele, I., & Palsson, B. Ø. (2010). What is flux balance analysis? *Nature Biotechnology*, *28*(3), 245–248. doi: 10.1038/nbt.1614

Oster, G. F., Perelson, A. S., & Katchalsky, A. (1971). Network thermodynamics. *Nature*, *234*, 393–399. doi: 10.1038/234393a0

Peusner, L. (1986). *Studies in network thermodynamics*. Amsterdam: Elsevier.

Qian, H., & Beard, D. A. (2005). Thermodynamics of stoichiometric biochemical networks in living systems far from equilibrium. *Biophysical Chemistry*, *114*(2–3), 213–220. doi: 10.1016/j.bpc.2004.12.001

Rapoport, T. A., & Heinrich, R. (1975). Mathematical analysis of multienzyme systems. I. Modelling of the glycolysis of human erythrocytes. *Biosystems*, *7*(1), 120–129. doi: 10.1016/0303-2647(75)90049-0

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLoS Computational Biology*, *9*(10), e1003285. doi: 10.1371/

journal.pcbi.1003285

Schuster, S., & Schuster, R. (1989). A generalization of the Wegscheider condition. Implications for properties of steady states and for quasi-steady-state approximation. *Journal of Mathematical Chemistry*, *3*(1), 25–42. doi: 10.1007/BF01171883

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, *22*(11), 1359–1366. doi: 10.1177/0956797611417632

The UniProt Consortium. (2025). UniProt: the universal protein knowledgebase in 2025. *Nucleic Acids Research*, *53*(D1), D609–D617. doi: 10.1093/nar/gkae1010

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., ... van Mulbregt, P. (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*, 261–272. doi: 10.1038/s41592-019-0686-2

Wegscheider, R. (1901). Über simultane gleichgewichte und die beziehungen zwischen thermodynamik und reactionskinetik homogener systeme. *Monatshefte für Chemie*, *22*(8), 849–906. doi: 10.1007/BF01517498

Zanella, A., Fermo, E., Bianchi, P., & Valentini, G. (2005). Red cell pyruvate kinase deficiency: molecular and clinical aspects. *British Journal of Haematology*, *130*(1), 11–25. doi: 10.1111/j.1365-2141.2005.05527.x